

Statistics: Probability & Hypothesis Testing



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DIRECTIONS

Answer each question. Show your work where applicable.

- 1 Classify as discrete or continuous:

Temperature in Fahrenheit

Answer: _____

- 2 Which type of error?

Concluding innocent when guilty

Answer: _____

- 3 Find the conditional probability:

$$\frac{P(A \cap B)}{P(B)} = \frac{0.6}{0.4}$$

Answer: _____

- 4 Contingency table is 2 times 3. Degrees of freedom = (rows-1)(cols-1).

$$df = (3 - 1)(2 - 1)$$

Answer: _____

- 5 Classify as discrete or continuous:

Weight of a newborn baby

Answer: _____

- 6 Which type of error?

Failing to reject a false null hypothesis

Answer: _____

- 7 Two cards drawn without replacement. P(both Kings)?

$$P(K_1 \text{ and } K_2) = \frac{4}{52} \cdot \frac{3}{51}$$

Answer: _____

- 8 Find the conditional probability:

$$\frac{P(A \cap B)}{P(B)} = \frac{0.7}{0.3}$$

Answer: _____



Answer Key & Solutions

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TEACHER NOTES

Topics: discrete vs. continuous variables, Type I/II errors, conditional probability, chi-square degrees of freedom, and dependent events.

- 1 Classify as discrete or continuous:

Temperature in Fahrenheit

= Continuous

"Temperature in Fahrenheit" is continuous because it takes on any value in a range

- 2 Which type of error?

Concluding innocent when guilty

= Type II Error

Type II error: concluding innocent when guilty.

- 3 Find the conditional probability:

$$\frac{P(A \cap B)}{P(B)} = \frac{0.6}{0.4}$$

= 1.50

Divide the joint probability by the marginal.

- 4 Contingency table is 2 times 3. Degrees of freedom = (rows-1)(cols-1).

$$df = (3 - 1)(2 - 1)$$

= 2

$df = 2 \times 1 = 2$.

- 5 Classify as discrete or continuous:

Weight of a newborn baby

= Continuous

"Weight of a newborn baby" is continuous because it takes on any value in a range

- 6 Which type of error?

Failing to reject a false null hypothesis

= Type II Error

Type II error: failing to reject a false null hypothesis.



- 7 Two cards drawn without replacement. P(both Kings)?

$$P(K_1 \text{ and } K_2) = \frac{4}{52} \cdot \frac{3}{51}$$

$$= \frac{12}{52 \cdot 51} = \frac{12}{2652}$$

$P(1st \text{ King}) \times P(2nd \text{ King} | 1st \text{ King})$. Result ≈ 0.0045 .

- 8 Find the conditional probability:

$$\frac{P(A \cap B)}{P(B)} = \frac{0.7}{0.3}$$

$$= 2.33$$

Divide the joint probability by the marginal.

